

Declan Chan

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Education

BASc in Engineering Science

2025–Present

University of Toronto

- Dean's Honours List (all semesters)

IB Diploma Programme

2021–2025

Maple High School

- IB Score: 45/45
- OSSD Top 6: 99.67%
- Director's Achievement Award

Technical Skills

Languages: Python, C, MATLAB, Bash

Design: SolidWorks, OnShape

Tools: Git, Linux, Slurm

Experience

Research Fellow

2026–Present

Autonomous Discovery of Alloys

- Designed experiments to evaluate the robustness of machine learning force fields.
- Benchmarked MLFF predictions against density functional theory relaxations.
- Performed large-scale experiments on Canada's Fir HPC cluster.

Space Systems Engineer

2026–Present

University of Toronto Aerospace Team

- Developed software for CubeSat attitude determination and control systems.
- Built Python modules for attitude dynamics and flight simulation.
- Simulated magnetic environments using a Helmholtz cage and MATLAB.

Mechanical Design Engineer

2025–2026

Robotics for Space Exploration

- Designed robotic arm components using CAD for 3D modelling.
- Performed finite element analysis in SolidWorks Simulation to evaluate designs.

Senior Instructor

2021–Present

Northern Karate Schools

- Yondan: Certified 4th Degree Black Belt.
- Gold medal at the Toronto International Tournament of Martial Arts Champions.
- Teach 100+ students each week through group and private lessons.

Projects

Machine Learning Force Field Attacks

2026

- Developed open-source software to generate gradient-based adversarial attacks on MLFFs.
- Built Python/PyTorch pipelines to process Materials Project and LiCOHPF datasets.
- Automated ASE relaxation, perturbations, and contour exploration workflows using Bash and Slurm.

GenAI Genesis Hackathon

2026

- Developed an interactive platform visualizing human disease progression.
- Implemented a RAG system trained on PubMed data via IBM watsonx.ai.
- Built an AI-powered 3D web application with React and three.js.

CIV102 Bridge Project

2025

- Optimized structural performance and prototyped a matboard bridge using Python.
- Top 1 in the cohort and 2nd in the class of 2025.

Astrophysics Research Project

2024–2025

- Conducted data analysis and literature review on Yarkovsky effect orbital perturbations.
- Modelled thermal-radiation orbital drift and trajectories of several near-Earth asteroids.

Generative Art

2021–2022

- Used the HashLips Art Engine to generate layered NFT artwork from image assets.
- Created and deployed NFTs on OpenSea.io using the Polygon Ethereum blockchain.

Extracurricular Achievements

Speedcubing Competitions

2022–2025

- Top 90 in Canada for 3x3x3 fewest moves.
- Volunteered as a judge for the World Cube Association.
- Club President who taught beginners and organized school-wide competitions.

Varsity Volleyball

2022–2025

- Team Captain in Grade 10 and 12.
- Tier 1 YRAA Championship junior varsity 2nd place.
- Tier 1 YRAA Championship senior varsity 2nd place.

Rep Volleyball

2019–2023

- Team Captain and 2× MVP.
- Silver medal in Canada's 2023 Youth Nationals.

Music Performance

2019–2023

- Theory and Instrumental: Level 10 RCM comprehensive certificate (First Class Honors).
- Level 10 RCM practical piano certificate (First Class Honors with Distinction).
- Level 8 RCM practical violin certificate (First Class Honors with Distinction).